

Chapter 05: Power Series Solutions of Linear Differential Equations

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Applied Ordinary Differential Equations (ENGR 213)

Textbook:

Advanced Engineering Mathematics

(7th Edition) by Dennis G. Zill — Chapter 5 (§5.1.1, §5.1.2)

1. Executive Overview & First-Principles Philosophy

In Chapters 2 and 3, our analytical methods for solving linear differential equations:

$$a_n(x) \frac{d^n y}{dx^n} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

relied entirely on two exceptional cases:

1. **Constant Coefficients:** Where the characteristic equation $ar^2 + br + c = 0$ yields closed-form exponential and sinusoidal eigenfunctions e^{rx} , $\cos(\beta x)$, $\sin(\beta x)$.
2. **Cauchy-Euler Equations:** Where coefficients are proportional to powers of x ($ax^2y'' + bxy' + cy = 0$), yielding polynomial eigenfunctions x^m .

However, in applied engineering physics, nature routinely produces differential equations with **variable coefficients** that are not Cauchy-Euler:

- **Airy's Equation:** $y'' - xy = 0$ (governing diffraction of light and quantum wavefunctions at turning points).
- **Legendre's Equation:** $(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$ (governing gravitational potential and spherical harmonics).
- **Bessel's Equation:** $x^2y'' + xy' + (x^2 - \nu^2)y = 0$ (governing heat dissipation in circular cylinders and vibrating drumheads).
- **Hermite's Equation:** $y'' - 2xy' + 2ny = 0$ (governing the quantum harmonic oscillator).

For such equations, **no finite combination of elementary functions (polynomials, exponentials, logarithms, or trigonometric functions) exists** to express the general solution.

The universal analytical engine for variable-coefficient linear differential equations is the **Power Series Method**:

- We represent the unknown function $y(x)$ as an infinite power series centered at a point x_0 :

$$y(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^n = c_0 + c_1(x - x_0) + c_2(x - x_0)^2 + \dots$$

- Substituting this series and its term-by-term derivatives into the ODE converts the differential relationship into a system of algebraic equations for the unknown coefficients c_n .
- The method systematically produces a **Recurrence Relation** that expresses all higher-order coefficients ($c_2, c_3, c_4, \dots$) entirely in terms of the two fundamental initial values:

$$c_0 = y(x_0) \quad \text{and} \quad c_1 = y'(x_0)$$

2. Mathematical Framework & Operational Engine

2.1 Review of Power Series (Zill §5.1.1)

A. Formal Definition & Convergence

An infinite series of the form:

$$\sum_{n=0}^{\infty} c_n (x - x_0)^n = c_0 + c_1 (x - x_0) + c_2 (x - x_0)^2 + \dots$$

is called a **power series centered at** x_0 . The constants c_n are the coefficients, and x_0 is the center of expansion. In ENGR 213, unless explicitly specified otherwise, we expand about the origin $x_0 = 0$:

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

B. Radius and Interval of Convergence

Every power series exhibits one of exactly three geometric convergence behaviors:

1. The series converges **only at its center** $x = x_0$ (Radius of convergence $R = 0$).
2. The series converges absolutely for **all real** x (Radius of convergence $R = \infty$).
3. There exists a finite positive number $R > 0$ such that the series converges absolutely for $|x - x_0| < R$ and diverges for $|x - x_0| > R$. The interval $(x_0 - R, x_0 + R)$ is the **interval of convergence**.

The radius of convergence R is determined using the **Ratio Test**:

$$L = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1} (x - x_0)^{n+1}}{c_n (x - x_0)^n} \right| = |x - x_0| \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|$$

- If $\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| = \rho$, then the series converges when $\rho |x - x_0| < 1$, which means:

$$R = \frac{1}{\rho} = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$$

C. Analytic Functions

A function $f(x)$ is said to be **analytic at a point** x_0 if it can be represented by a power series in $(x - x_0)$ with a strictly positive radius of convergence $R > 0$:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$$

- **Every polynomial** is analytic everywhere on $(-\infty, \infty)$ with $R = \infty$.
- **Elementary transcendental functions** e^x , $\sin x$, $\cos x$, $\sinh x$, $\cosh x$ are analytic everywhere on $(-\infty, \infty)$ with $R = \infty$.
- **A rational function** $f(x) = \frac{P(x)}{Q(x)}$ (where P and Q are polynomials with no common factors) is analytic at every point x_0 where the denominator is non-zero: $Q(x_0) \neq 0$.

D. Operational Calculus on Power Series

Within its open interval of convergence $|x - x_0| < R$, a power series defines a continuous function with derivatives of all orders. We perform calculus **term-by-term**:

1. First Derivative:

$$y'(x) = \frac{d}{dx} \sum_{n=0}^{\infty} c_n x^n = \sum_{n=1}^{\infty} n c_n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + \dots$$

(Note: The lower limit shifts from $n = 0$ to $n = 1$ because the derivative of the constant c_0 is zero).

2. Second Derivative:

$$y''(x) = \frac{d^2}{dx^2} \sum_{n=0}^{\infty} c_n x^n = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + \dots$$

(Note: The lower limit shifts to $n = 2$ because the first two derivative terms are zero).

3. The Identity Property (Vanishing Principle):

If a power series is identically zero for all x in an open interval:

$$\sum_{n=0}^{\infty} d_n (x - x_0)^n = 0 \quad \forall x \in (x_0 - R, x_0 + R) \iff d_n = 0 \quad \text{for every } n = 0, 1, 2, \dots$$

Every single coefficient must vanish independently!

2.2 The Mechanics of Index Shifting (The Summation Engine)

In differential equations, substituting series produces multiple sums carrying different powers of x :

$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+1}$$

To combine these into a single summation and equate the net coefficient to zero, **all sums must be re-indexed so they share the exact identical generic power x^k** .

The 3-Step Re-Indexing Law:

1. **Define the common exponent:** Set the dummy variable k equal to the current exponent of x .

2. **Solve for original index n :** Express n in terms of k , and substitute into all coefficients.
3. **Recalculate the lower limit:** Evaluate k at the starting value of n .

Original Sum	Exponent Substitution	New Sum in x^k	Starting Value
$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}$	Let $k = n - 2 \implies n = k + 2$	$\sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k$	$n = 2 \implies k = 0$
$\sum_{n=1}^{\infty} n c_n x^{n-1}$	Let $k = n - 1 \implies n = k + 1$	$\sum_{k=0}^{\infty} (k+1)c_{k+1} x^k$	$n = 1 \implies k = 0$
$\sum_{n=1}^{\infty} n c_n x^n$	Let $k = n \implies n = k$	$\sum_{k=1}^{\infty} k c_k x^k$	$n = 1 \implies k = 1$
$\sum_{n=0}^{\infty} c_n x^{n+1}$	Let $k = n + 1 \implies n = k - 1$	$\sum_{k=1}^{\infty} c_{k-1} x^k$	$n = 0 \implies k = 1$
$\sum_{n=0}^{\infty} c_n x^{n+2}$	Let $k = n + 2 \implies n = k - 2$	$\sum_{k=2}^{\infty} c_{k-2} x^k$	$n = 0 \implies k = 2$

Unrolling / Peeling Off Lower Terms

If one sum starts at $k = 0$ and another starts at $k = 1$, they cannot be combined directly under a single $\sum$ sign. We must **peel off** the $k = 0$ term from the first sum:

$$\sum_{k=0}^{\infty} A_k x^k + \sum_{k=1}^{\infty} B_k x^k = (A_0 x^0) + \sum_{k=1}^{\infty} [A_k + B_k] x^k = 0$$

Then by the Identity Property:

$$A_0 = 0 \quad \text{and} \quad A_k + B_k = 0 \quad \text{for all } k \geq 1$$

2.3 Ordinary Points vs. Singular Points (Zill §5.1.2)

Consider the general second-order linear differential equation with variable coefficients:

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

Dividing by the leading coefficient $a_2(x)$ puts the equation into **standard normal form**:

$$y'' + P(x)y' + Q(x)y = 0, \quad \text{where } P(x) = \frac{a_1(x)}{a_2(x)}, \quad Q(x) = \frac{a_0(x)}{a_2(x)}$$

A. Definitions of Ordinary and Singular Points

- **Ordinary Point:** A point x_0 is called an **ordinary point** of the differential equation if both coefficient functions $P(x)$ and $Q(x)$ are **analytic at x_0** .
 - When $a_2(x), a_1(x), a_0(x)$ are polynomials with no common factors, x_0 is an ordinary point **if and only if** $a_2(x_0) \neq 0$.
- **Singular Point:** If either $P(x)$ or $Q(x)$ fails to be analytic at x_0 , the point x_0 is called a **singular point**.
 - For polynomial coefficients, singular points are precisely the **roots of the leading coefficient**: $a_2(x) = 0$.

B. Existence Theorem for Power Series Solutions (Theorem 5.1.1)

Theorem: If x_0 is an ordinary point of the differential equation:

$$y'' + P(x)y' + Q(x)y = 0$$

then there **always exist two linearly independent solutions** in the form of a power series centered at x_0 :

$$y(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^n$$

The general solution is given by:

$$y(x) = c_0 y_1(x) + c_1 y_2(x)$$

where c_0 and c_1 are arbitrary constants representing $y(x_0)$ and $y'(x_0)$, respectively.

C. Minimum Guaranteed Radius of Convergence (Distance in Complex Plane)

The radius of convergence R of each series solution $y_1(x)$ and $y_2(x)$ is **at least as large as the distance from x_0 to the nearest singular point (real or complex) of the differential equation:**

$$R \geq \min_{z_{\text{sing}}} |x_0 - z_{\text{sing}}|$$

bound for the radius of convergence. For example, the complex numbers $1 \pm 2i$ are singular points of $(x^2 - 2x + 5)y'' + xy' - y = 0$, but since $x = 0$ is an ordinary point of the equation, Theorem 5.1.1 guarantees that we can find two power series solutions centered at 0. That is, the solutions look like $\sum_{n=0}^{\infty} c_n x^n$, and, moreover, we know without finding these solutions that each series must converge *at least* for $|x| < \sqrt{4}$, where $R = \sqrt{4}$ is the distance in the complex plane from 0 to either of the points $1 + 2i$ or $1 - 2i$. See **FIGURE 5.1.2**. However, the differential equation has a solution that converges for much larger values of x ; indeed, this series is valid on the interval $(-\infty, \infty)$ because it can

Figure 5.1.2: Geometric determination of the minimum radius of convergence R in the complex plane. For $(x^2 - 2x + 5)y'' + xy' - y = 0$, singular points occur at roots of $z^2 - 2z + 5 = 0 \implies z = 1 \pm 2i$. The distance from ordinary point $x_0 = 0$ to $1 \pm 2i$ is $R = \sqrt{1^2 + 2^2} = \sqrt{5}$.

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Example 1

: $(x^2 + 4)y'' + 2xy' + y = 0$ expanded about $x_0 = 0$.

- Singular points: $x^2 + 4 = 0 \implies z = \pm 2i$.
- Distance from 0 to $\pm 2i$: $R \geq |0 - 2i| = \sqrt{0^2 + 2^2} = 2$.
- Both series solutions are guaranteed to converge at least on $(-2, 2)$.

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Example 2

: $(x^2 - 2x + 2)y'' + y = 0$ expanded about $x_0 = 0$.

- Singular points: $z^2 - 2z + 2 = 0 \implies z = \frac{2 \pm \sqrt{4-8}}{2} = 1 \pm i$.
- Distance from 0 to $1 + i$: $R \geq |1 + i| = \sqrt{1^2 + 1^2} = \sqrt{2}$.

3. The 7-Step Universal Power Series Algorithm

To solve any linear 2nd-order ODE about an ordinary point $x_0 = 0$ on Concordia exams, execute these 7 baby steps:

7-Step Universal Power Series Algorithm

Step 1: Check Ordinary Point & Determine Guaranteed Radius R

Verify $a_2(x_0) \neq 0$. Find roots of $a_2(z) = 0$.

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Step 2: State Standard Series Definitions

$y = \sum c_n x^n$, $y' = \sum n c_n x^{n-1}$, $y'' = \sum n(n-1) c_n x^{n-2}$

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Step 3: Substitute Series into the ODE

Multiply polynomial coefficients into the sums.

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Step 4: Shift Indices to Generic Power x^k

Transform every sum so the exponent of x is k .

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Step 5: Align Lower Limits by Peeling Off Excess Terms

Evaluate $k = 0$ (and $k = 1$ if needed) terms separately.

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Step 6: Formulate the Recurrence Relation

Set individual peeled terms to 0, and net bracket = 0.

Solve for highest index coefficient: $c_{k+2} = f(c_k, c_{k-1})$.

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Step 7: Compute Coefficients Iteratively & Assemble $y(x)$

Calculate $c_2, c_3, c_4, \dots$ in terms of c_0 and c_1 .

Group into $y(x) = c_0 y_1(x) + c_1 y_2(x)$.

4. Comprehensive Step-by-Step Problem Walkthroughs

4.1 Problem 1: Airy's Differential Equation ($y'' - xy = 0$) (Zill §5.1 Archetype)

Problem Statement: Find the general power series solution centered at the ordinary point $x_0 = 0$ for Airy's equation:

$$y'' - xy = 0$$

Determine the first 4 non-zero terms of each fundamental linearly independent solution $y_1(x)$ and $y_2(x)$, and specify the guaranteed radius of convergence.

Step 1: Check Ordinary Point and Guaranteed Radius

The leading coefficient is $a_2(x) = 1 \neq 0$ for all $x \in \mathbb{R}$. There are **no singular points** anywhere in the complex plane ($R = \infty$). The solution is guaranteed to converge for all $x \in (-\infty, \infty)$.

Step 2: Set Up Series Representations

$$y(x) = \sum_{n=0}^{\infty} c_n x^n, \quad y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}, \quad y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

Step 3: Substitute Series into Airy's Equation

$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n = 0$$

Distribute x into the second summation:

$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Step 4: Shift Indices to Common Exponent x^k

- **First Sum:** Let $k = n - 2 \implies n = k + 2$. When $n = 2$, $k = 2 - 2 = 0$.

$$\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$$

- **Second Sum:** Let $k = n + 1 \implies n = k - 1$. When $n = 0$, $k = 0 + 1 = 1$.

$$\sum_{k=1}^{\infty} c_{k-1} x^k$$

Equation becomes:

$$\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$$

Step 5: Align Starting Indices by Peeling Off $k = 0$

The first sum starts at $k = 0$, while the second sum starts at $k = 1$. We peel off the $k = 0$ term from the first sum:

- For $k = 0$: $(0 + 2)(0 + 1)c_{0+2}x^0 = 2c_2$. Now combine the remaining sums from $k = 1$ to ∞ :

$$2c_2 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - c_{k-1}] x^k = 0$$

Step 6: Formulate the Recurrence Relation

By the Identity Property, every coefficient must equal zero independently:

1. **Constant term** (x^0):

$$2c_2 = 0 \implies c_2 = 0$$

2. **General terms** (x^k for $k \geq 1$):

$$(k+2)(k+1)c_{k+2} - c_{k-1} = 0$$

Isolating the higher index coefficient c_{k+2} :

$$c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)} \quad \text{for } k = 1, 2, 3, 4, \dots$$

Crucial Observation

: Each coefficient c_{k+2} is linked to c_{k-1} —which is **3 steps back** ($k+2 - (k-1) = 3$). This means the coefficients split naturally into **three independent families**:

- Family 1 (depends on c_0): $c_0 \rightarrow c_3 \rightarrow c_6 \rightarrow c_9 \rightarrow \dots$ (indices multiples of 3).
- Family 2 (depends on c_1): $c_1 \rightarrow c_4 \rightarrow c_7 \rightarrow c_{10} \rightarrow \dots$ (indices $\equiv 1 \pmod{3}$).
- Family 3 (depends on c_2): $c_2 \rightarrow c_5 \rightarrow c_8 \rightarrow c_{11} \rightarrow \dots$ (indices $\equiv 2 \pmod{3}$). Since $c_2 = 0$, **all coefficients in Family 3 are identically zero!**

Step 7: Iterate the Recurrence Relation Step-by-Step

- **For** $k = 1$:

$$c_3 = \frac{c_0}{(3)(2)} = \frac{c_0}{6}$$

- **For** $k = 2$:

$$c_4 = \frac{c_1}{(4)(3)} = \frac{c_1}{12}$$

- **For** $k = 3$:

$$c_5 = \frac{c_2}{(5)(4)} = \frac{0}{20} = 0$$

- **For** $k = 4$:

$$c_6 = \frac{c_3}{(6)(5)} = \frac{c_0/6}{30} = \frac{c_0}{180}$$

- For $k = 5$:

$$c_7 = \frac{c_4}{(7)(6)} = \frac{c_1/12}{42} = \frac{c_1}{504}$$

- For $k = 6$:

$$c_8 = \frac{c_5}{(8)(7)} = \frac{0}{56} = 0$$

- For $k = 7$:

$$c_9 = \frac{c_6}{(9)(8)} = \frac{c_0/180}{72} = \frac{c_0}{12960}$$

- For $k = 8$:

$$c_{10} = \frac{c_7}{(10)(9)} = \frac{c_1/504}{90} = \frac{c_1}{45360}$$

Step 8: Assemble the General Solution

Substitute the calculated coefficients back into $y(x) = \sum_{n=0}^{\infty} c_n x^n$:

$$y(x) = c_0 + c_1 x + 0x^2 + \frac{c_0}{6}x^3 + \frac{c_1}{12}x^4 + 0x^5 + \frac{c_0}{180}x^6 + \frac{c_1}{504}x^7 + 0x^8 + \frac{c_0}{12960}x^9 + \dots$$

Group terms factoring out c_0 and c_1 :

$$y(x) = c_0 \left(1 + \frac{x^3}{6} + \frac{x^6}{180} + \frac{x^9}{12960} + \dots \right) + c_1 \left(x + \frac{x^4}{12} + \frac{x^7}{504} + \frac{x^{10}}{45360} + \dots \right)$$

$$y(x) = c_0 y_1(x) + c_1 y_2(x)$$

$y_1(x)$ and $y_2(x)$ shown in **FIGURE 5.1.3** were obtained with the aid of a numerical solver using the fourth-order Runge–Kutta method with a step size of $\Delta x = 0.1$.

Figure 5.1.3: Graphical behavior of Airy solutions $y_1(x)$ and $y_2(x)$. For $x < 0$, the solutions exhibit oscillatory behavior (like $\sin$ and $\cos$); for $x > 0$, they exhibit exponential growth and decay.

4.2 Problem 2: Variable Coefficients with Polynomial Termination ($y'' - 2xy' + 2y = 0$)

Problem Statement: Find the general power series solution about $x_0 = 0$ for Hermite's differential equation with parameter $n = 1$:

$$y'' - 2xy' + 2y = 0$$

Step 1: Check Ordinary Point and Guaranteed Radius

Leading coefficient is $a_2(x) = 1 \neq 0$. The point $x_0 = 0$ is an ordinary point, and $R = \infty$.

Step 2: Substitute Series

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} - 2x \sum_{n=1}^{\infty} n c_n x^{n-1} + 2 \sum_{n=0}^{\infty} c_n x^n = 0$$

Distribute $2x$ and 2 into the sums:

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} - \sum_{n=1}^{\infty} 2n c_n x^n + \sum_{n=0}^{\infty} 2c_n x^n = 0$$

Step 3: Shift Indices to Generic Power x^k

- **First sum:** Let $k = n - 2 \implies n = k + 2$. Starts at $k = 0$:

$$\sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k$$

- **Second sum:** Let $k = n \implies n = k$. Starts at $k = 1$:

$$\sum_{k=1}^{\infty} 2k c_k x^k$$

- **Third sum:** Let $k = n \implies n = k$. Starts at $k = 0$:

$$\sum_{k=0}^{\infty} 2c_k x^k$$

Step 4: Align Lower Limits by Peeling Off $k = 0$

Peel off $k = 0$ from the first and third sums:

- First sum ($k = 0$): $(2)(1)c_2 = 2c_2$.
- Third sum ($k = 0$): $2c_0$. Remaining summation from $k = 1$ to ∞ :

$$(2c_2 + 2c_0) + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - 2kc_k + 2c_k] x^k = 0$$

Factor the bracket:

$$(2c_2 + 2c_0) + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - 2(k-1)c_k] x^k = 0$$

Step 5: Derive the Recurrence Relation

1. **Constant term** (x^0):

$$2c_2 + 2c_0 = 0 \implies c_2 = -c_0$$

2. **General term** (x^k for $k \geq 1$):

$$(k+2)(k+1)c_{k+2} - 2(k-1)c_k = 0 \implies c_{k+2} = \frac{2(k-1)}{(k+2)(k+1)}c_k, \quad k \geq 1$$

Crucial Observation

: Here c_{k+2} depends on c_k (step size 2). Even indices depend on c_0 , odd indices depend on c_1 .

Step 6: Iterate Coefficients

• **Odd Coefficients** (c_1 family):

◦ For $k = 1$:

$$c_3 = \frac{2(1-1)}{(3)(2)}c_1 = \frac{0}{6}c_1 = 0!$$

◦ For $k = 3$:

$$c_5 = \frac{2(3-1)}{(5)(4)}c_3 = \frac{4}{20}(0) = 0$$

◦ Because $c_3 = 0$, **all subsequent odd coefficients vanish identically**: $c_3 = c_5 = c_7 = \dots = 0$.

◦ **The second solution terminates into a finite 1st-degree polynomial**:

$$y_2(x) = c_1 x$$

• **Even Coefficients** (c_0 family):

◦ $c_2 = -c_0$

◦ For $k = 2$:

$$c_4 = \frac{2(2-1)}{(4)(3)}c_2 = \frac{2}{12}(-c_0) = -\frac{1}{6}c_0$$

◦ For $k = 4$:

$$c_6 = \frac{2(4-1)}{(6)(5)}c_4 = \frac{6}{30}\left(-\frac{1}{6}c_0\right) = -\frac{1}{30}c_0$$

◦ For $k = 6$:

$$c_8 = \frac{2(6-1)}{(8)(7)}c_6 = \frac{10}{56}\left(-\frac{1}{30}c_0\right) = -\frac{1}{168}c_0$$

Step 7: Assemble the General Solution

$$y(x) = c_0 \left(1 - x^2 - \frac{1}{6}x^4 - \frac{1}{30}x^6 - \frac{1}{168}x^8 - \dots \right) + c_1 x$$

Phenomenon of Polynomial Solutions

: When differential equations arise from physical eigenvalue problems (like the Hermite or Legendre equations), for integer parameters n , one of the series branches terminates into a **finite polynomial** (the Hermite polynomial $H_1(x) = 2x$), which represents a physically bounded, normalizable quantum wavefunction!

5. Critical Concordia Exam Traps & Red Flags

- **⚠ Trap 1: Forgetting Complex Roots When Finding Radius R :** In $(x^2 + 9)y'' + xy = 0$, students often look for real roots, find none, and incorrectly conclude $R = \infty$.

The Law

: You must solve $z^2 + 9 = 0$ in the **complex plane** $\implies z = \pm 3i$. The distance from $x_0 = 0$ to $3i$ is $|0 - 3i| = 3$. The radius of convergence is $R = 3$, not ∞ !

- **⚠ Trap 2: Premature Summation Merging:** Never combine sums before their lower summation limits match! If $\sum_{k=0}^{\infty} A_k x^k + \sum_{k=1}^{\infty} B_k x^k = 0$, you cannot write $\sum_{k=0}^{\infty} (A_k + B_k) x^k = 0$ because B_0 does not exist! You must peel off A_0 first: $A_0 + \sum_{k=1}^{\infty} (A_k + B_k) x^k = 0$.
- **⚠ Trap 3: Miscalculating the Recurrence Step:** In Airy's equation $c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}$, the index step is $(k+2) - (k-1) = 3$. Do not compute c_2 from c_0 ; c_2 comes from the peeled constant term $2c_2 = 0 \implies c_2 = 0$. c_3 comes from c_0 , c_4 comes from c_1 , and c_5 comes from c_2 .
- **⚠ Trap 4: Omitting the Independent Constants c_0 and c_1 :** A second-order ODE requires **two arbitrary constants**. Never set $c_0 = 1$ and $c_1 = 1$ during the derivation. Keep c_0 and c_1 as algebraic parameters until the very end, and clearly factor them out as $y(x) = c_0 y_1(x) + c_1 y_2(x)$.